EXPERIMENT 16

AIM:

To perform basic data exploration and visualisation of the given dataset using Python.

CODE:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the dataset
df = pd.read_csv("/content/student_eda_dataset.csv")
print(df)

# Display first five rows
print(df.head())

# Distribution plot of Marks with KDE
sns.displot(df.Marks, kde=True)
plt.show()

# Distribution plot of Marks without KDE
sns.displot(df.Marks, kde=False)
plt.show()
```

```
# Joint plot of Study Hours and Marks
sns.jointplot(x=df.Study_Hours, y=df.Marks)
plt.show()

# Regression plot
sns.jointplot(x=df.Study_Hours, y=df.Marks, kind="reg")
plt.show()

# Hexagonal joint plot
sns.jointplot(x=df.Study_Hours, y=df.Marks, kind="hex")
plt.show()

# Pair plot of numerical features
sns.pairplot(
    df[['Study_Hours', 'Attendance', 'Marks', 'Assignment_Score']]
)
plt.show()

# Pair plot based on Study Mode
sns.pairplot(
    df[['Study_Hours', 'Attendance', 'Marks',
        'Assignment_Score', 'Study_Mode']],
    hue='Study_Mode'
)
```

```
plt.show()

# Pair plot based on Gender
sns.pairplot(
    df[['Study_Hours', 'Attendance', 'Marks',
        'Assignment_Score', 'Gender']],
    hue='Gender'
)
plt.show()

# Correlation heatmap
sns.heatmap(
    df[['Study_Hours', 'Attendance', 'Marks',
        'Assignment_Score']].corr(),
    annot=True
)
plt.show()

# Box plot of Marks
sns.boxplot(y=df.Marks)
plt.show()

# Box plot of Study Hours
sns.boxplot(y=df.Study_Hours)
plt.show()
```

```
# Count plot of Department
sns.countplot(x=df.Department)
plt.show()

# Count plot of Gender
sns.countplot(x=df.Gender)
plt.show()

# Pie chart of Gender
df.Gender.value_counts().plot(
    kind='pie', autopct='%1.1f%%'
)
plt.ylabel('')
plt.show()

# Bar chart of Gender
df.Gender.value_counts().plot(kind='bar')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.show()

# Count plot of Pass status
sns.countplot(x=df.Passed)
plt.show()
```

```
# Count plot of Study Mode
sns.countplot(x=df.Study_Mode)
plt.show()

# Gender versus Pass status
sns.countplot(x=df.Gender, hue=df.Passed)
plt.show()

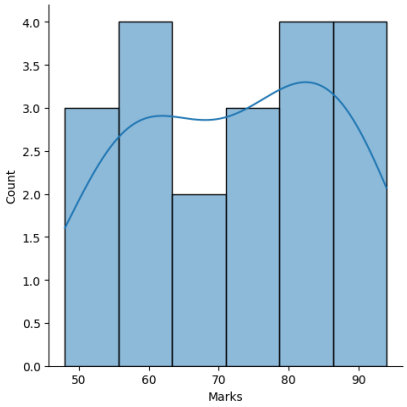
# Department versus Pass status
sns.countplot(x=df.Department, hue=df.Passed)
plt.show()

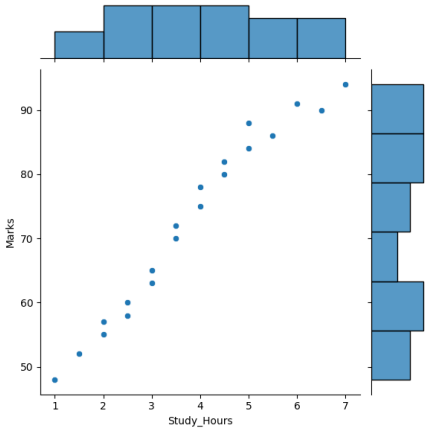
# Department distribution for Online study mode
sns.countplot(
    x=df[df.Study_Mode == 'Online']['Department']
)
plt.show()
```

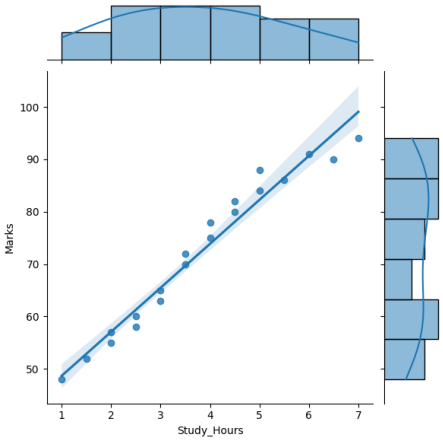
OUTPUT:

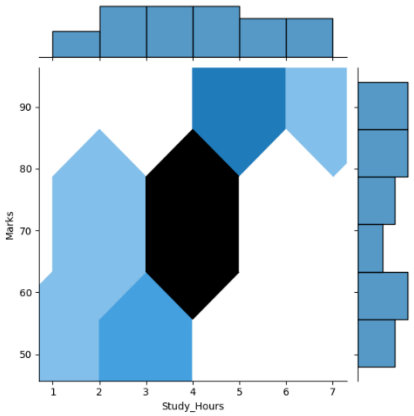
	Student_ID	Gender	Department	Study_Mode	Study_Hours	Attendance	Marks	Assignment_Score	Passed
0	101	Male	CSE	Online	2.5	72	58	60	No
1	102	Female	ECE	Offline	4.0	85	78	80	Yes
2	103	Male	CSE	Online	3.0	78	65	68	Yes
3	104	Female	IT	Offline	5.0	92	88	90	Yes
4	105	Male	ECE	Online	1.5	68	52	55	No
5	106	Female	CSE	Offline	6.0	95	91	93	Yes
6	107	Male	IT	Online	3.5	80	70	72	Yes
7	108	Female	ECE	Offline	4.5	88	82	85	Yes
8	109	Male	CSE	Online	2.0	70	55	58	No
9	110	Female	IT	Offline	5.5	90	86	88	Yes
10	111	Male	ECE	Offline	7.0	96	94	96	Yes
11	112	Female	CSE	Online	3.0	76	63	65	Yes
12	113	Male	IT	Offline	4.0	84	75	78	Yes
13	114	Female	ECE	Online	2.5	73	60	62	Yes
14	115	Male	CSE	Offline	6.5	94	90	92	Yes
15	116	Female	IT	Online	1.0	65	48	50	No
16	117	Male	ECE	Offline	5.0	89	84	87	Yes
17	118	Female	CSE	Online	3.5	82	72	75	Yes
18	119	Male	IT	Offline	2.0	71	57	59	No
19	120	Female	ECE	Online	4.5	87	80	83	Yes

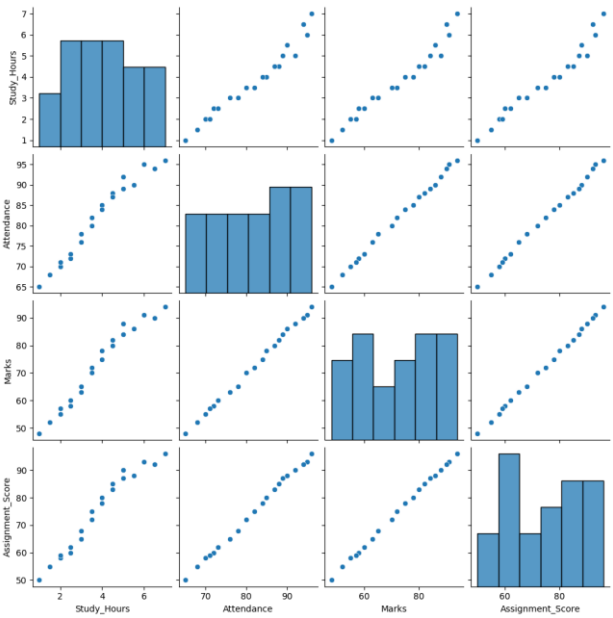
	Student_ID	Gender	Department	Study_Mode	Study_Hours	Attendance	Marks	Assignment_Score	Passed
0	101	Male	CSE	Online	2.5	72	58	60	No
1	102	Female	ECE	Offline	4.0	85	78	80	Yes
2	103	Male	CSE	Online	3.0	78	65	68	Yes
3	104	Female	IT	Offline	5.0	92	88	90	Yes
4	105	Male	ECE	Online	1.5	68	52	55	No

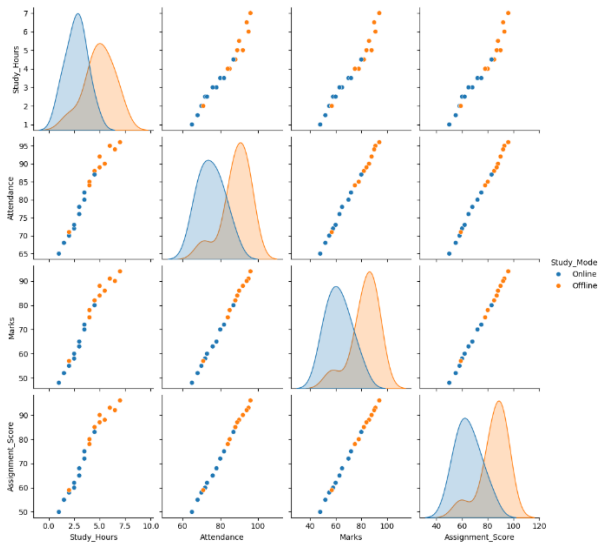

Count
Marks

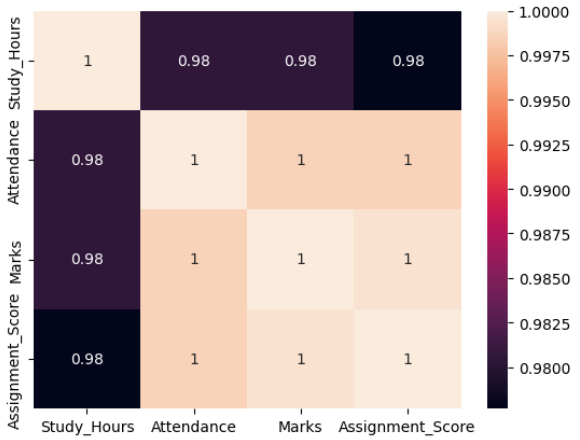

Marks
Study_Hours

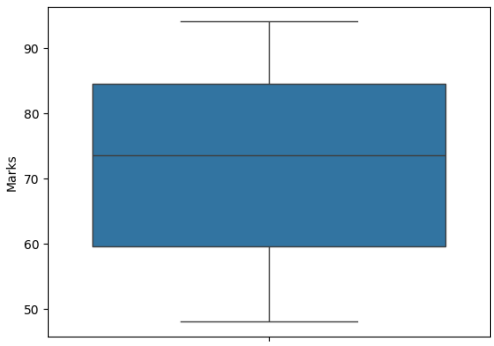

Marks
Study_Hours

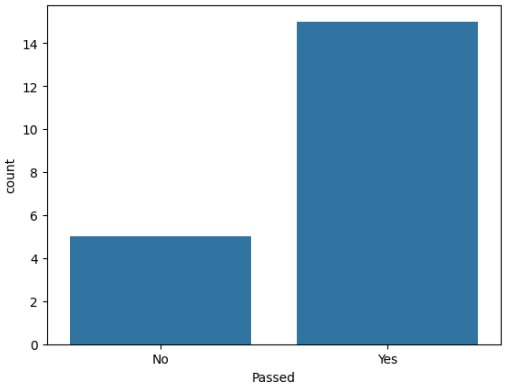

Marks
Study_Hours

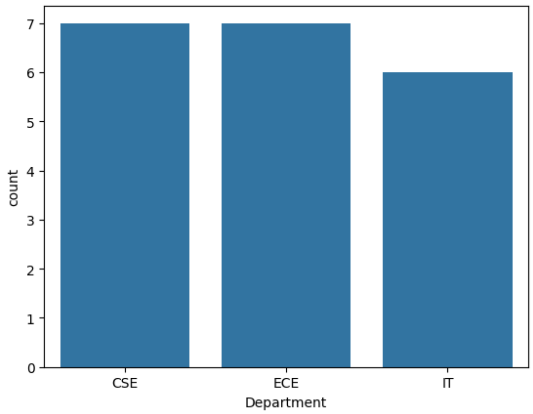

Study_Hours
Attendance
Marks
Assignment_Score

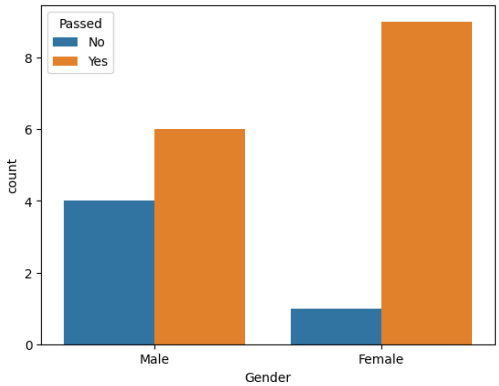

Study_Hours
Attendance
Marks
Assignment_Score
Study_Mode
Online
Offline

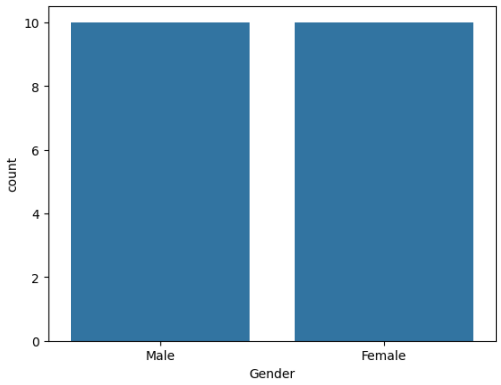

Study_Hours
Attendance
Marks
Assignment_Score
1
0.98
0.98
0.98
0.98
1
1
1
0.98
1
1
1
0.98
1
1
1
1.0000
0.9975
0.9950
0.9925
0.9900
0.9875
0.9850
0.9825
0.9800

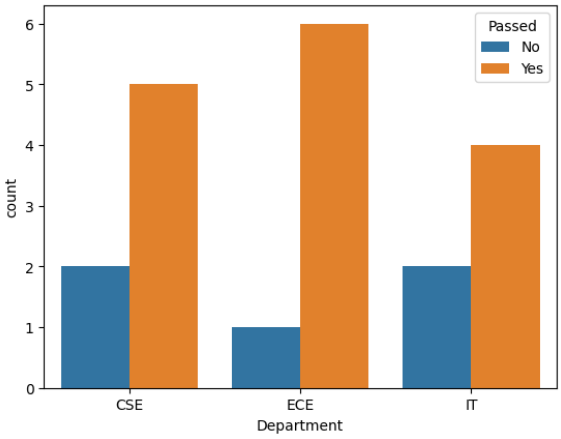

Marks
90
80
70
60
50

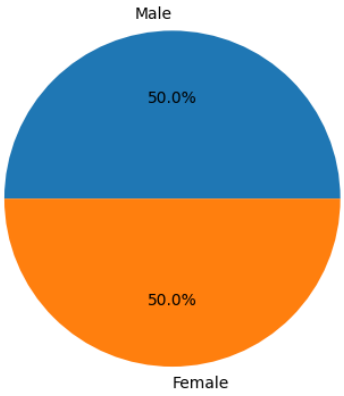

count
14
12
10
8
6
4
2
0
No
Yes
Passed


count
Department
CSE
ECE
IT
0
1
2
3
4
5
6
7


Passed
No
Yes
count
Gender
Male
Female
0
2
4
6
8


count
Gender
Male
Female
0
2
4
6
8
10


Passed
No
Yes
count
Department
CSE
ECE
IT
0
1
2
3
4
5
6


Male
50.0%
50.0%
Female

RESULT:

The given dataset was successfully loaded, explored, and visualised using Python, providing a better understanding of its data characteristics and relationships.